

Update 10-march-2026

Let me explain **very simply** what we are trying to achieve with all this setup.

1. The Goal

We are trying to **run your AI marketplace system (PickPic / Monetchat)** on your server so people can use it online.

This system includes several components working together:

- Web application (Next.js marketplace)
- Database (PostgreSQL)
- Vector search engine (Qdrant)
- AI model (Ollama / Llama)
- Background services

All of these need to run **reliably on your server**.

2. The Problem Without Docker

If we install everything manually on the server:

Example:

- Install Node.js
- Install PostgreSQL
- Install Qdrant
- Install Ollama
- Configure environment variables
- Manage ports

- Manage updates

Problems:

- Very hard to maintain
 - Version conflicts
 - Difficult to move to another server
 - If something breaks it takes hours to fix
 - Scaling becomes difficult
-

3. What Docker Solves

Docker packages each part of your system into **containers**.

Think of a container like a **mini isolated computer**.

Example:

Your server will run containers like this:

```
Server
├── Container 1 → Web App (Next.js)
├── Container 2 → PostgreSQL Database
├── Container 3 → Qdrant Vector Database
├── Container 4 → Ollama AI
└── Container 5 → Redis (optional)
```

Each container runs independently.

Benefits:

- Easy to start/stop
- Easy to update
- No conflicts
- Very stable



- Easy to move to another server

4. What We Are Actually Doing Right Now

We are preparing your server so it can run containers.

Steps we are doing:

Step 1

Enable **virtualization**

Why?

Docker needs virtualization to run containers efficiently.

You already did this.

Step 2

Install **Docker Engine**

This is the software that manages containers.

Step 3

Install **Docker Compose**

Docker Compose lets us start **all services with one command**.

Example:

Instead of starting 5 things separately, we run:

```
docker compose up -d
```

And it starts:

- database
- vector search
- AI model
- backend
- frontend

All automatically.

5. What Will Happen After Docker Is Ready

Once Docker is installed we will deploy your system like this:

```
docker compose up -d
```

Then your server will automatically run:



Users will then access it from:

`http://your-server-ip`

6. The Big Picture

What we are building is basically:

Your own AI marketplace platform

Features:

- buyers search with AI
- upload image to find products
- voice search
- sellers upload items
- AI understands the request
- vector search finds best results

And it runs **entirely on your own server**.

7. Why This Is Powerful

Because you are not depending on:

- OpenAI
- external APIs

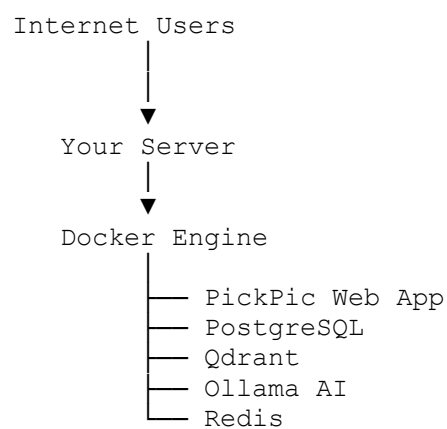
Everything runs **locally on your infrastructure**.

That means:

- cheaper
- faster
- more control
- scalable

8. The Final Architecture

Your server will look like this:



9. Where You Are Right Now

You have already done **the hardest parts**:

- ✓ Server created
- ✓ Windows Server installed

- ✓ Ollama working locally
- ✓ Qdrant working
- ✓ Vector data indexed
- ✓ Virtualization enabled

Now we are just **making deployment clean and stable with Docker.**

10. Important Question You Should Actually Ask

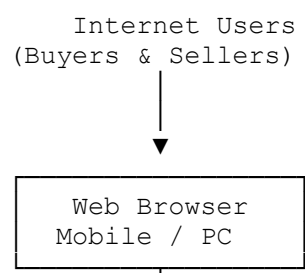
Before we go further we must answer something important:

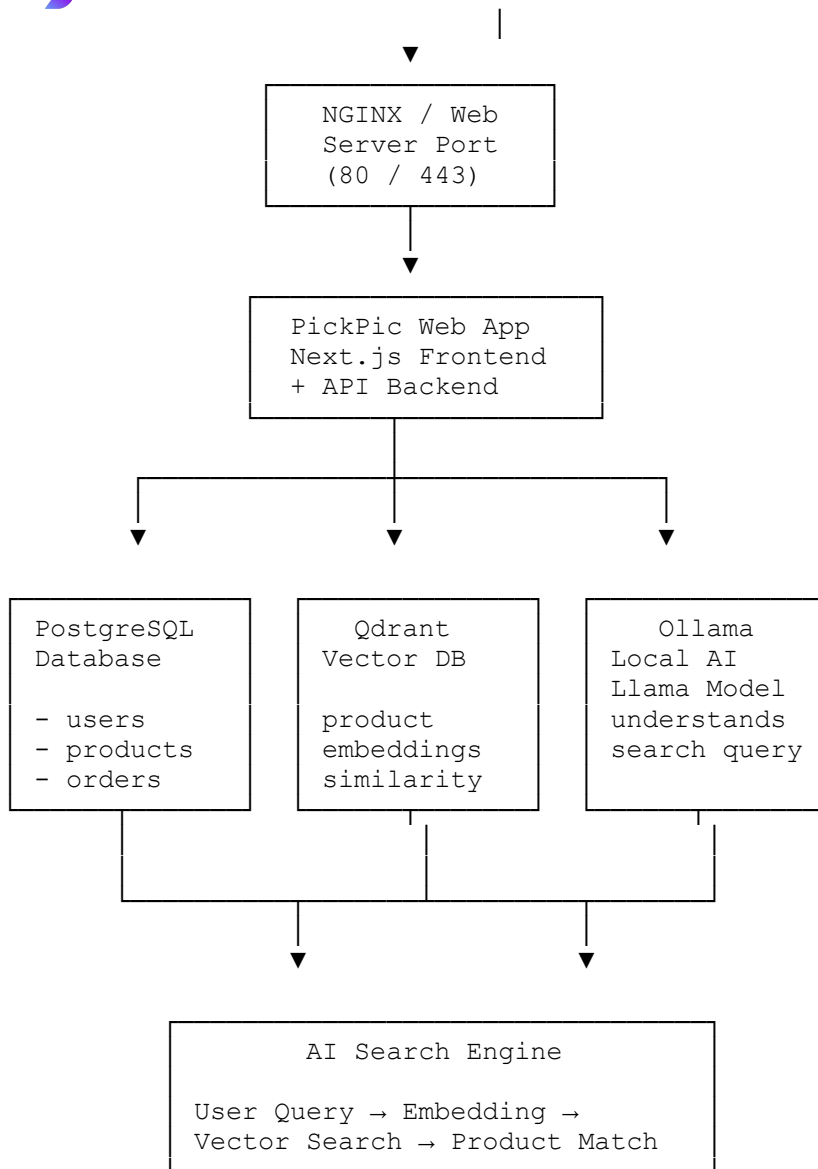
Should we run Docker on Windows Server... or switch to Linux?

Because:

Linux servers run Docker **much better and faster.**

Here is the **simple architecture diagram** of what we are building and deploying on your server.





How a User Search Actually Works

Example:

User types:

"black leather office chair"

Process:

User Query



Next.js Backend

Ollama AI
(converts text to embedding)

Qdrant Vector Database
(searches similar products)

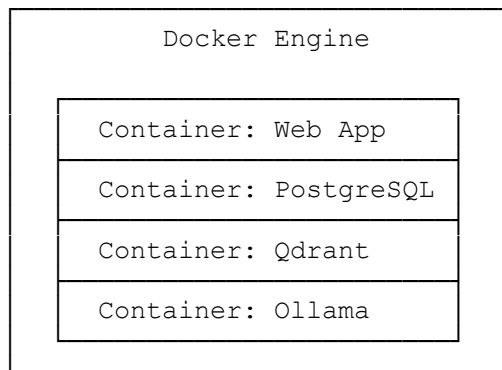
PostgreSQL
(fetch product details)

Results returned to user

Where Docker Fits In

Docker simply **runs each component safely**.

Your Server



The Final Result

When everything is deployed:

Users open:



They will see:

- AI chat marketplace
- product search
- image search
- seller listings
- AI recommendations

All running **from your own server**.